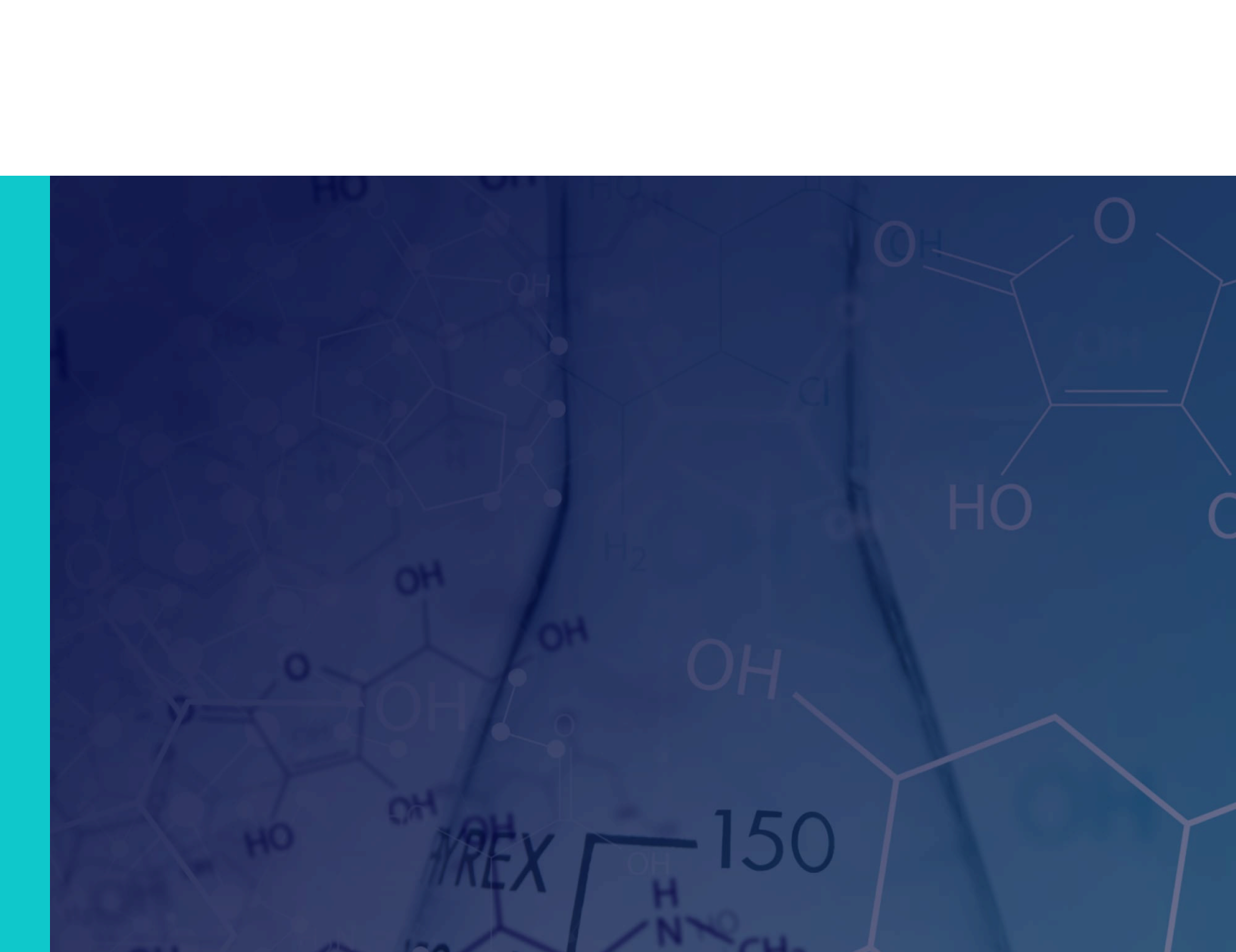




McGill

CHEM 110

Fall 2024, Chapter 2 Notes



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2. Electron Configuration and Chemical Periodicity

2.1 Orbital Filling Diagrams

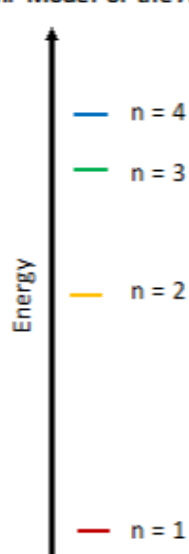
2.1.1

Relative Energies of Atomic Orbitals

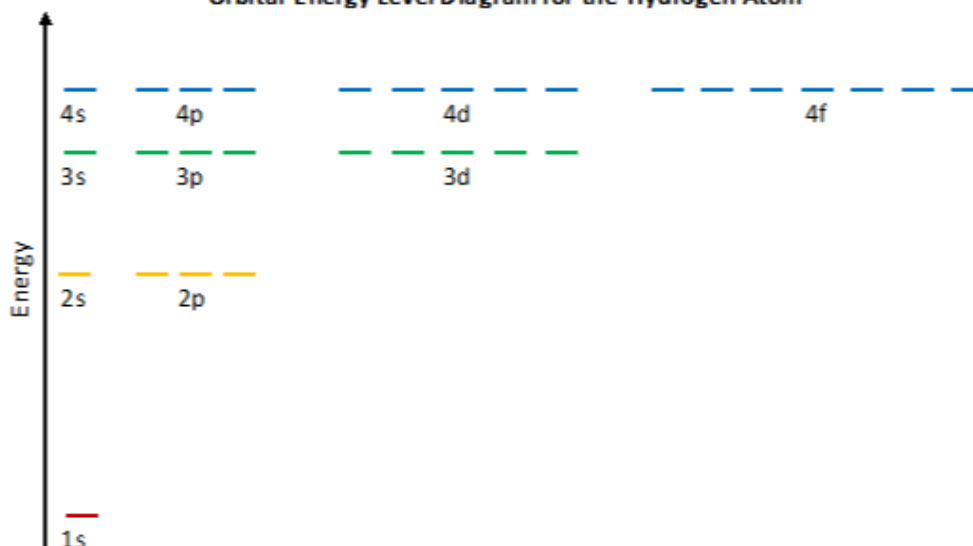
One-Electron Species

- For a one-electron species like the hydrogen atom, there is no electron-electron repulsion. Therefore, all the subshells of the same n are **degenerate**.
- Orbitals at the **same energy level** are called **degenerate**.

Bohr Model of the Atom

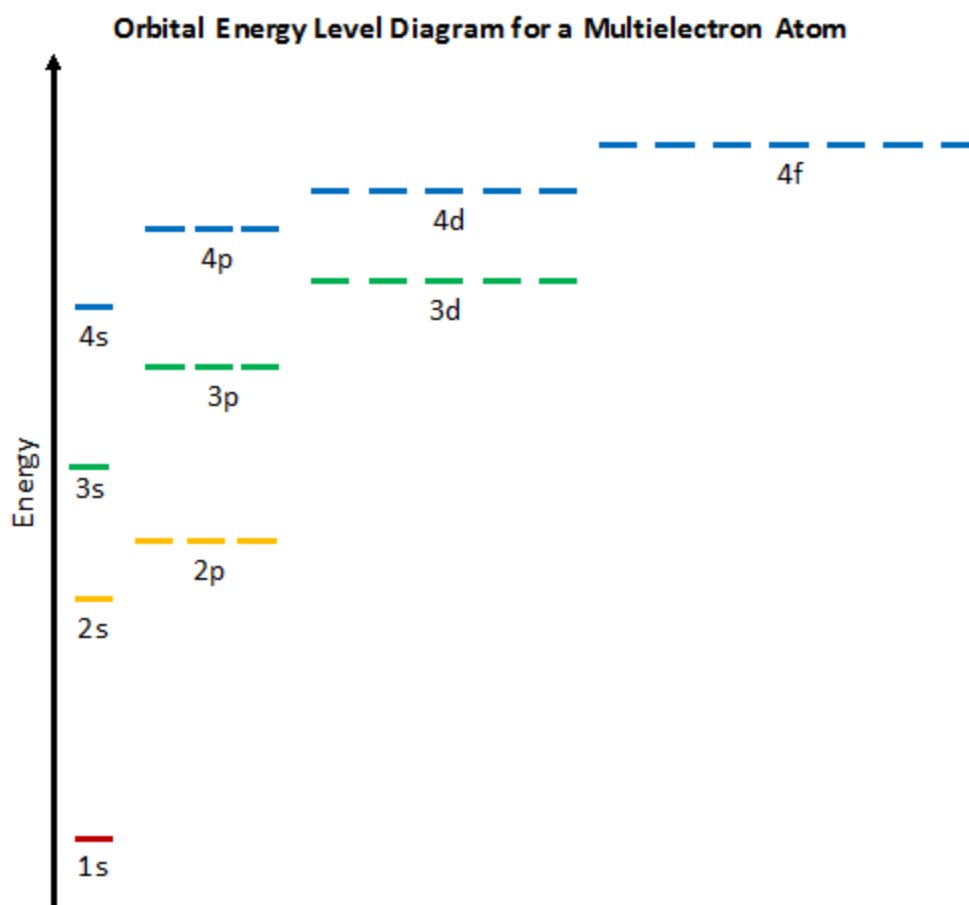


Orbital Energy Level Diagram for the Hydrogen Atom



Multi-Electron Species

- With multi-electron species, there are interactions between electrons (electron-electron repulsion)
- The energy level diagram of the orbitals looks like this:





WIZE CONCEPT

- S subshells hold a maximum of _____ electrons
- P subshells hold a maximum of _____ electrons
- D subshells hold a maximum of _____ electrons
- F subshells hold a maximum of _____ electrons

The total number of orbitals in a given shell is given by : n^2

Example:

If $n=2$, what is the total number of orbitals in that shell?

- _____, circle the _____ orbitals that have $n=2$ above

What if we wanted to know the maximum number of electrons we could have if $n=2$?

- We already found that total number of orbitals for this shell (_____)
- What is the maximum number of electrons we could have in each orbital? _____
- Therefore the maximum number of electrons possible when $n=2$ is _____

Rules for Orbital Filling

1) Aufbau Principle

Electrons will always occupy the **lowest available energy level first**.

1s																1s
2s													2p			
3s													3p			
4s					3d							4p				
5s					4d							5p				
6s					5d							6p				
7s					6d							7p				

						4f						
						5f						

Diagram illustrating the filling order of atomic orbitals (Aufbau principle):

- 1s
- 2s 2p
- 3s 3p 3d
- 4s 4p 4d 4f
- 5s 5p 5d 5f ...
- 6s 6p 6d ...

Red arrows indicate the sequence of filling: 1s → 2s → 2p → 3s → 3p → 4s → 3d → 4p → 5s → 4d → 5p → 6s → 4f → 5d → 6p → 7s → 5f → 6d → 7p → 8s → 6f → 7d → 8p → 9s → 7f → 8d → 9p → 10s → 8f → 9d → 10p → 11s → 9f → 10d → 11p → 12s → 10f → 11d → 12p → 13s → 11f → 12d → 13p → 14s → 12f → 13d → 14p → 15s → 13f → 14d → 15p → 16s → 14f → 15d → 16p → 17s → 15f → 16d → 17p → 18s → 16f → 17d → 18p → 19s → 17f → 18d → 19p → 20s → 18f → 19d → 20p → 21s → 19f → 20d → 21p → 22s → 20f → 21d → 22p → 23s → 21f → 22d → 23p → 24s → 22f → 23d → 24p → 25s → 23f → 24d → 25p → 26s → 24f → 25d → 26p → 27s → 25f → 26d → 27p → 28s → 26f → 27d → 28p → 29s → 27f → 28d → 29p → 30s → 28f → 29d → 30p → 31s → 29f → 30d → 31p → 32s → 30f → 31d → 32p → 33s → 31f → 32d → 33p → 34s → 32f → 33d → 34p → 35s → 33f → 34d → 35p → 36s → 34f → 35d → 36p → 37s → 35f → 36d → 37p → 38s → 36f → 37d → 38p → 39s → 37f → 38d → 39p → 40s → 38f → 39d → 40p → 41s → 39f → 40d → 41p → 42s → 40f → 41d → 42p → 43s → 41f → 42d → 43p → 44s → 42f → 43d → 44p → 45s → 43f → 44d → 45p → 46s → 44f → 45d → 46p → 47s → 45f → 46d → 47p → 48s → 46f → 47d → 48p → 49s → 47f → 48d → 49p → 50s → 48f → 49d → 50p → 51s → 49f → 50d → 51p → 52s → 50f → 51d → 52p → 53s → 51f → 52d → 53p → 54s → 52f → 53d → 54p → 55s → 53f → 54d → 55p → 56s → 54f → 55d → 56p → 57s → 55f → 56d → 57p → 58s → 56f → 57d → 58p → 59s → 57f → 58d → 59p → 60s → 58f → 59d → 60p → 61s → 59f → 60d → 61p → 62s → 60f → 61d → 62p → 63s → 61f → 62d → 63p → 64s → 62f → 63d → 64p → 65s → 63f → 64d → 65p → 66s → 64f → 65d → 66p → 67s → 65f → 66d → 67p → 68s → 66f → 67d → 68p → 69s → 67f → 68d → 69p → 70s → 68f → 69d → 70p → 71s → 69f → 70d → 71p → 72s → 70f → 71d → 72p → 73s → 71f → 72d → 73p → 74s → 72f → 73d → 74p → 75s → 73f → 74d → 75p → 76s → 74f → 75d → 76p → 77s → 75f → 76d → 77p → 78s → 76f → 77d → 78p → 79s → 77f → 78d → 79p → 80s → 78f → 79d → 80p → 81s → 79f → 80d → 81p → 82s → 80f → 81d → 82p → 83s → 81f → 82d → 83p → 84s → 82f → 83d → 84p → 85s → 83f → 84d → 85p → 86s → 84f → 85d → 86p → 87s → 85f → 86d → 87p → 88s → 86f → 87d → 88p → 89s → 87f → 88d → 89p → 90s → 88f → 89d → 90p → 91s → 89f → 90d → 91p → 92s → 90f → 91d → 92p → 93s → 91f → 92d → 93p → 94s → 92f → 93d → 94p → 95s → 93f → 94d → 95p → 96s → 94f → 95d → 96p → 97s → 95f → 96d → 97p → 98s → 96f → 97d → 98p → 99s → 97f → 98d → 99p → 100s → 98f → 99d → 100p → 101s → 99f → 100d → 101p → 102s → 100f → 101d → 102p → 103s → 101f → 102d → 103p → 104s → 102f → 103d → 104p → 105s → 103f → 104d → 105p → 106s → 104f → 105d → 106p → 107s → 105f → 106d → 107p → 108s → 106f → 107d → 108p → 109s → 107f → 108d → 109p → 110s → 108f → 109d → 110p → 111s → 109f → 110d → 111p → 112s → 110f → 111d → 112p → 113s → 111f → 112d → 113p → 114s → 112f → 113d → 114p → 115s → 113f → 114d → 115p → 116s → 114f → 115d → 116p → 117s → 115f → 116d → 117p → 118s → 116f → 117d → 118p → 119s → 117f → 118d → 119p → 120s → 118f → 119d → 120p → 121s → 119f → 120d → 121p → 122s → 120f → 121d → 122p → 123s → 121f → 122d → 123p → 124s → 122f → 123d → 124p → 125s → 123f → 124d → 125p → 126s → 124f → 125d → 126p → 127s → 125f → 126d → 127p → 128s → 126f → 127d → 128p → 129s → 127f → 128d → 129p → 130s → 128f → 129d → 130p → 131s → 129f → 130d → 131p → 132s → 130f → 131d → 132p → 133s → 131f → 132d → 133p → 134s → 132f → 133d → 134p → 135s → 133f → 134d → 135p → 136s → 134f → 135d → 136p → 137s → 135f → 136d → 137p → 138s → 136f → 137d → 138p → 139s → 137f → 138d → 139p → 140s → 138f → 139d → 140p → 141s → 139f → 140d → 141p → 142s → 140f → 141d → 142p → 143s → 141f → 142d → 143p → 144s → 142f → 143d → 144p → 145s → 143f → 144d → 145p → 146s → 144f → 145d → 146p → 147s → 145f → 146d → 147p → 148s → 146f → 147d → 148p → 149s → 147f → 148d → 149p → 150s → 148f → 149d → 150p → 151s → 149f → 150d → 151p → 152s → 150f → 151d → 152p → 153s → 151f → 152d → 153p → 154s → 152f → 153d → 154p → 155s → 153f → 154d → 155p → 156s → 154f → 155d → 156p → 157s → 155f → 156d → 157p → 158s → 156f → 157d → 158p → 159s → 157f → 158d → 159p → 160s → 158f → 159d → 160p → 161s → 159f → 160d → 161p → 162s → 160f → 161d → 162p → 163s → 161f → 162d → 163p → 164s → 162f → 163d → 164p → 165s → 163f → 164d → 165p → 166s → 164f → 165d → 166p → 167s → 165f → 166d → 167p → 168s → 166f → 167d → 168p → 169s → 167f → 168d → 169p → 170s → 168f → 169d → 170p → 171s → 169f → 170d → 171p → 172s → 170f → 171d → 172p → 173s → 171f → 172d → 173p → 174s → 172f → 173d → 174p → 175s → 173f → 174d → 175p → 176s → 174f → 175d → 176p → 177s → 175f → 176d → 177p → 178s → 176f → 177d → 178p → 179s → 177f → 178d → 179p → 180s → 178f → 179d

! WATCH OUT!

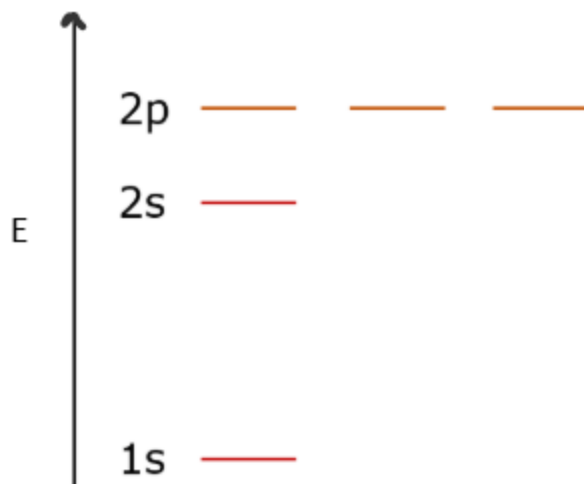
Memorize the order of orbital filling (or just familiarize yourself enough with the periodic table to know the order!): 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p...

2) Hund's Rule

Due to electron-electron repulsion, electrons will **fill orbitals of the same energy singly before pairing up**

Electrons don't want to be next to each other unless they have to be!

Example: Fill out the following orbital diagram for C



3) Pauli Exclusion Principle

No two electrons in an atom will have the same set of **4 quantum numbers**.

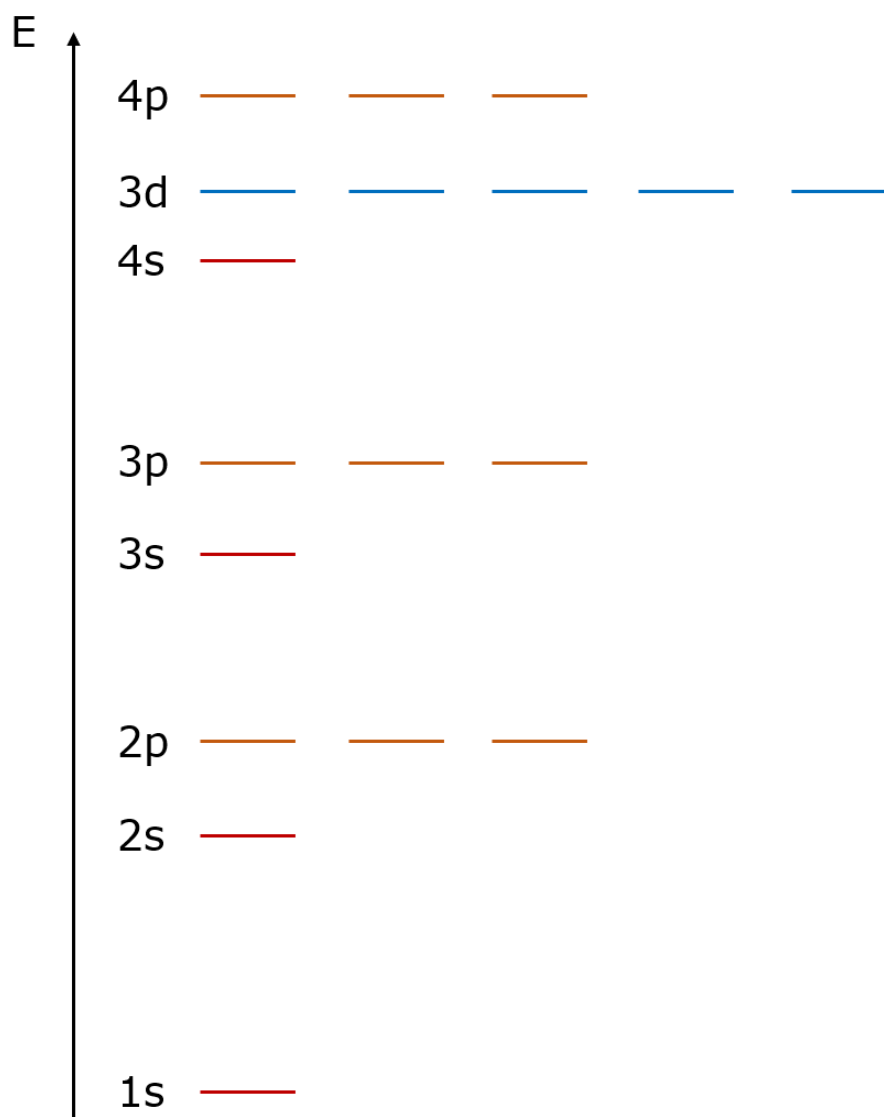
Quantum Number	Values	Interpretation
m_s (the spin quantum number)	-1/2 or +1/2	An electron behaves like a magnet that has one of two possible orientations, aligned either with the magnetic field or against it

Example: Orbital Filling Diagrams

Draw the orbital diagram for oxygen.

Example: Orbital Filling Diagrams

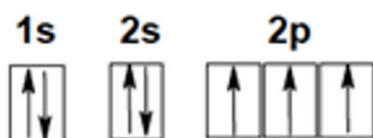
Complete the atomic orbital diagram below for a neutral calcium atom.



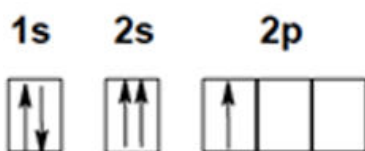
Practice: Orbital Filling Diagrams

Which of the orbital diagrams gives the correct electron configuration for an atom of boron, in the ground state?

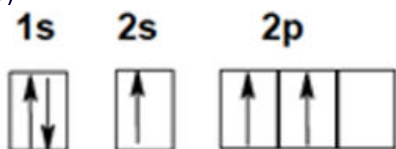
A)

☐

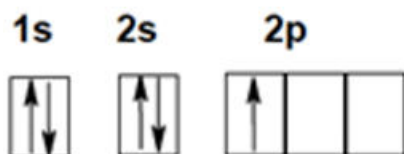
B)

☐

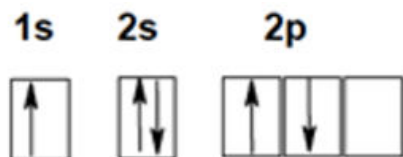
C)

☐

D)

☐

E)

☐

Practice: Number of Orbitals

How many different atomic orbitals exist where $n = 3$?

3

☐

6

☐

9

☐

12, which

☐

18

☐

2.2 Electronic Configuration for Atoms

2.2.1

Electron Configurations of Atoms

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

- lable blocks
- describe the n number for each block

Simple Electron Configurations

Li:

C:

Ne:

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Shorthand Notations

Write the name of the **previous noble gas** and then fill in the rest of the electron configuration.

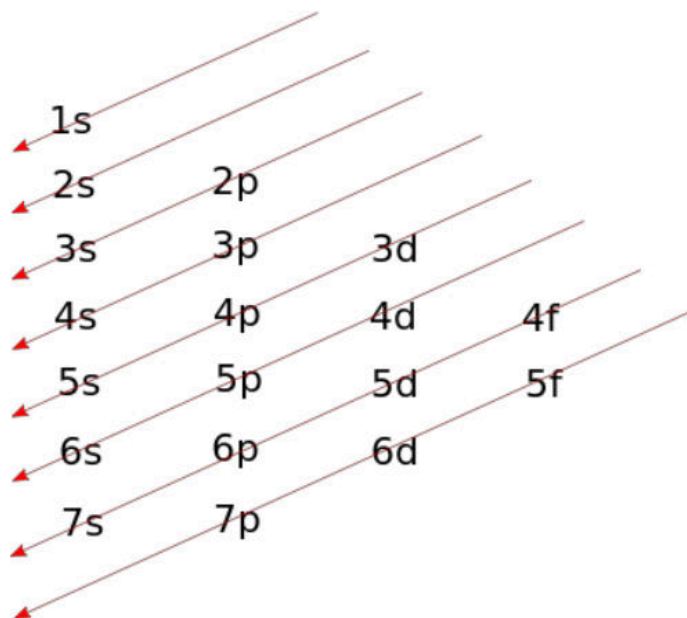
O:

Fe:

Ge:

Electron Configurations Cheatsheet

1 H 1.008 ←1s→																		2 He 4.003																					
3 Li 6.941 ←2s→		4 Be 9.012																5 B 10.81 2p→		6 C 12.011 2p→		7 N 14.007 2p→		8 O 15.999 2p→		9 F 18.998 2p→		10 Ne 20.180 2p→											
11 Na 22.990 ←3s→		12 Mg 24.305 ←3s→																13 Al 26.982 3p→		14 Si 28.085 3p→		15 P 30.974 3p→		16 S 32.06 3p→		17 Cl 35.45 3p→		18 Ar 39.948 3p→											
19 K 39.098 ←4s→		20 Ca 40.078 ←4s→		21 Sc 44.956 3d→		22 Ti 47.867 3d→		23 V 50.942 3d→		24 Cr 51.996 3d→		25 Mn 54.938 3d→		26 Fe 55.845 3d→		27 Co 58.933 3d→		28 Ni 58.693 3d→		29 Cu 63.546 3d→		30 Zn 65.38 3d→		31 Ga 69.723 4p→		32 Ge 72.630 4p→		33 As 74.922 4p→		34 Se 78.97 4p→		35 Br 79.904 4p→		36 Kr 83.798 4p→					
37 Rb 85.468 ←5s→		38 Sr 87.62 ←5s→		39 Y 88.906 4d→		40 Zr 91.224 4d→		41 Nb 92.906 4d→		42 Mo 95.95 4d→		43 Tc [97] 4d→		44 Ru 101.07 4d→		45 Rh 102.906 4d→		46 Pd 106.42 4d→		47 Ag 107.868 4d→		48 Cd 112.414 4d→		49 In 114.818 5p→		50 Sn 118.710 5p→		51 Sb 121.760 5p→		52 Te 127.60 5p→		53 I 126.904 5p→		54 Xe 131.293 5p→					
55 Cs 132.905 ←6s→		56 Ba 137.327 ←6s→		*		57 - 70		71 La 138.905 5d→		72 Ce 140.12 5d→		73 Pr 140.908 5d→		74 Nd 144.242 5d→		75 Pm [145] 5d→		76 Sm 150.36 5d→		77 Eu 151.964 5d→		78 Gd 157.25 5d→		79 Tb 158.925 5d→		80 Dy 162.500 5d→		81 Ho 164.930 5d→		82 Er 167.259 5d→		83 Tm 168.934 5d→		84 Yb 173.045 5d→					
87 Fr [223] ←7s→		88 Ra [226] ←7s→		**		89 - 102		103 Db [261] 6d→		104 Rf [261] 6d→		105 Db [262] 6d→		106 Sg [266] 6d→		107 Bh [264] 6d→		108 Hs [277] 6d→		109 Mt [268] 6d→		110 Ds [271] 6d→		111 Rg [272] 6d→		112 Cn [285] 6d→		113 Nh [284] 7p→		114 Fl [289] 7p→		115 Mc [289] 7p→		116 Lv [293] 7p→		117 Ts [293] 7p→		118 Og [294] 7p→	
*Lanthanide series				4f→		57 La 138.905		58 Ce 140.118		59 Pr 140.908		60 Nd 144.242		61 Pm [145]		62 Sm 150.36		63 Eu 151.964		64 Gd 157.25		65 Tb 158.925		66 Dy 162.500		67 Ho 164.930		68 Er 167.259		69 Tm 168.934		70 Yb 173.045							
**Actinide series				5f→		89 Ac [227]		90 Th 232.038		91 Pa 231.036		92 U 238.029		93 Np [237]		94 Pu [244]		95 Am [243]		96 Cm [247]		97 Bk [247]		98 Cf [251]		99 Es [252]		100 Fm [257]		101 Md [258]		102 No [259]							



Electron Configuration Exceptions

Transition metals have their valence electrons in the d subshell.

It is a lot **more favourable** (and stable) to have **d shell either half full or totally full instead of just partially full**.

	Cr:		Cu:
Expected:	 4s 3d [Ar] 4s ² 3d ⁴	 4s 3d [Ar] 4s ² 3d ⁹	
Seen:	 4s 3d [Ar] 4s ¹ 3d ⁵	 4s 3d [Ar] 4s ¹ 3d ¹⁰	

i WIZE TIP

The other exceptions that follow this same trend are: Mo, Ag, and Au! (Note they are in the same groups that Cr and Cu are in!)

Example: Writing Out Electron Configurations

Part 1:

Write the full electronic configurations for the following elements:

P:

Se:

Part 2:

Write the short electron configurations for the following elements:

Zn:

Zr:

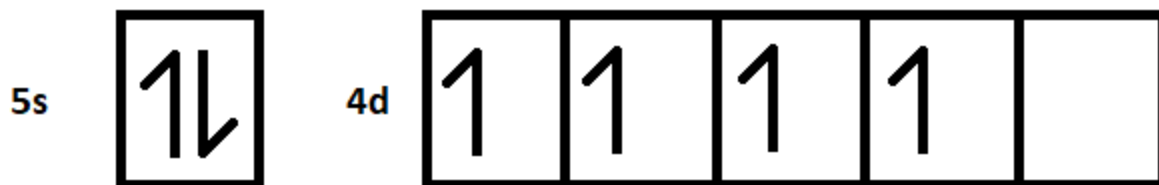
Group→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period	The Periodic Table of the Elements																	
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

Example: Condensed Electron Configuration

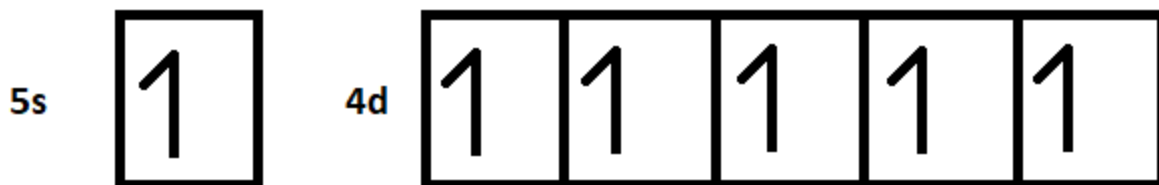
What is the condensed electronic configuration for Mo?

Group→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period	The Periodic Table of the Elements																	
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

We ignore everything before the previous noble gas (Kr), which means we are only working with the 5s and 4d blocks. Let's fill them up:



But there is something special about Mo. It is in group 6. This group tends to fill (or in this case half-fill) the d-block at the expense of the s block.



Therefore, the electronic configuration is $[\text{Kr}]5s^14d^5$.

Practice: Identify the Neutral Element

What neutral element is represented in the example below?

$1s^2 2s^2 2p^6 3s^2 3p^5$ became $[Ne] 3s^2 3p^5$

Al

☐

Ne

☐

P

☐

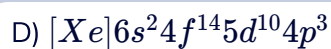
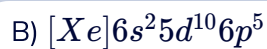
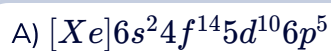
Cl

☐

Practice: Electron Configuration

The ground state electronic configuration for At is

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	



2.2.8 Practice

Practice: Impossible Electron Configuration

Which of the following electronic configurations is not possible?

Group → 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

↓ Period

The Periodic Table of the Elements

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

Lanthanides

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Actinides

89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
----------	----------	----------	---------	----------	----------	----------	----------	----------	----------	----------	-----------	-----------	-----------	-----------

[Ar]4s²4d⁶



1s²2s²2p⁶3s²3p⁶4s¹



[Kr]5s¹



1s²2s²2p⁶3s²3p⁶4s²3d⁸



2.3 Electron Configurations for Ions

2.3.1

Electron Configuration of Ions of Main Group Elements

We will now look at how to write the electron configurations for ions!

Writing the electron configurations for anions (negative ions) are simple.

1. Write out the electron configuration for the **neutral element**
2. Fill in the desired number of electrons by **adding electron(s) to the highest energy subshell**

Writing the electron configurations for cations (positive ions) usually confuses students more, but there is a simple thing to remember to get these questions right every time!

1. Write out the electron configuration for the **neutral element**
2. Remove the desired number of electrons by **first removing electrons from the highest n level and highest energy subshell!**

WIZE CONCEPT

The higher the n level, the higher the energy.

Example: 2s has a higher energy than 1s

In order of **increasing energy for the subshells**, we write:

s < p < d < f

Example: 2p has a higher energy than 2s

F: $1s^2 2s^2 2p^5$

[He] $2s^2 2p^5$



F⁻: $1s^2 2s^2 2p^6$

[Ne]



What is the electron configuration for F⁺?

i WIZE TIP

The most stable ions of atoms are **isoelectronic** with the noble gases or have filled shells. **Isoelectronic species** have the same number of electrons and the same electron configuration. The term "isoelectronic" is commonly seen on exams.

Example: F⁻ is isoelectronic with Ne!

Electron Configuration of Ions of Transition Elements

The same rules that we just learned apply for transition elements as well. Note that valence electrons are only removed, never core electrons.

! WATCH OUT!

Transition metals can lose both “n” and “n-1” valence electrons, but “n” electrons are always lost first! Let's take a look!

Fe: [Ar] 4s²3d⁶



Fe¹⁺: [Ar] 4s¹3d⁶



Fe²⁺: [Ar] 3d⁶



Fe³⁺: [Ar] 3d⁵



i WIZE TIP

Always write out the **electron configuration for the neutral atom first**.
Then write out the **electron configuration for the charged element**.

This will help you avoid mistakes on the exam!

Example: Isoelectronic Species

Two atoms or ions are said to be isoelectronic if the electron configuration is the same for both species. Which of the following pairs are isoelectronic?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

i) N^+ and C

ii) C and B^+

Example: Writing Electron Configurations of Ions

Write the electronic configuration for the following species:

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

a) N^+

b) Ni^{2+}

Practice: Electron Configurations of Ions

For which of the atom/ions below, is the given electron configuration correct? (select all that apply)

	Species	Electron Configurations
i.	Cu	$[\text{Ar}] 4s^2 3d^9$
ii.	Ti^{2+}	$[\text{Ar}] 4s^2$
iii.	As^{5+}	$[\text{Ar}] 4s^2 3d^8$

i.

☐

ii.

☐

iii.

☐

none of the above

☐

2.4 Diamagnetic vs Paramagnetic Electron Configurations

2.4.1

Diamagnetic vs Paramagnetic

You may see these terms come up when talking about electron configurations.

Diamagnetic - all electrons are **paired**

Paramagnetic - all electrons are **not paired**

Are the following electron configurations diamagnetic or paramagnetic?

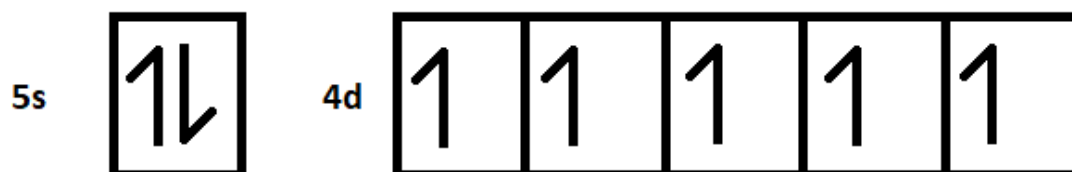
Cd

[Kr] 5s² 4d¹⁰



Tc

[Kr] 5s² 4d⁵



Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Consider the electron configuration for Na

Is it diamagnetic or paramagnetic? _____

Consider the electron configuration for Si

Is it diamagnetic or paramagnetic? _____

Consider the electron configuration for Be

 WIZE CONCEPT

Diamagnetic-all electrons are **paired**. If something is diamagnetic it will also be repelled from externally produced magnetic fields.

Paramagnetic-One or more electrons are left **unpaired**. If something is paramagnetic it will be attracted to an externally produced magnetic field.

Example: Paramagnetic and Diamagnetic Species

Write the electronic configuration of the following species, and label them as **D** diamagnetic or **P** paramagnetic

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

a) Sc^{2+}

b) Cr^{2+}

Practice: Diamagnetic

Which of the following are diamagnetic? (select all that apply in your answer)

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Cr

☐

Co

☐

Ru

☐

None of the above

☐

2.5 Effective Nuclear Charge (Zeff)

2.5.1

Effective Nuclear Charge (Zeff)

To understand effective nuclear charge (Zeff), let's consider a concert:

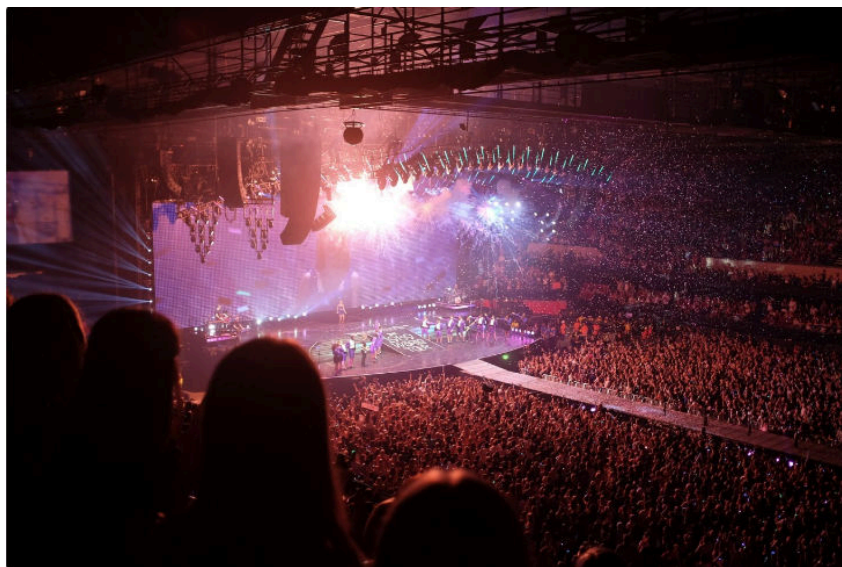


Photo by The Come Up Show / CC BY

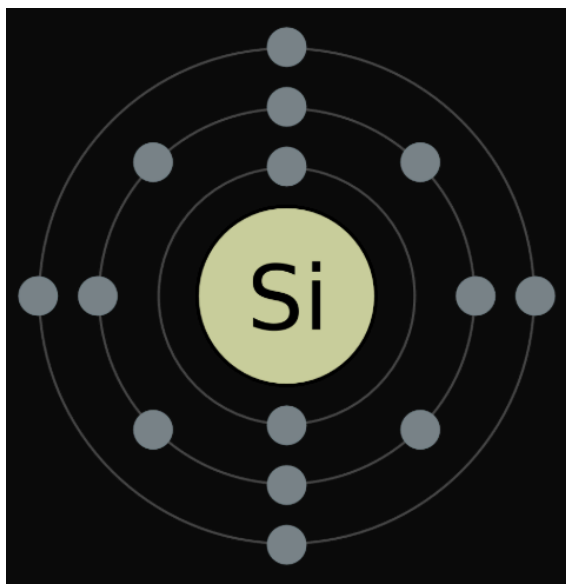
- People closer to the stage at a concert are going to be more into it. The music is louder and they really get a close connection with the artist. People in the highest rows in the stadium get less of that connection with the artist performing.
- This helps to explain the concept of nuclear shielding. **Inner electrons** (or people sitting in the front rows in our example) **shield the outer electrons** (or people sitting in the highest rows) from the **attractive force of the nucleus** (or artist in our example)

Effective Nuclear Charge

This is the **nuclear charge that is "felt"** by a valence electron.

Core Electrons Vs Valence Electrons

Label the **core** and **valence electrons** in the diagram below:



- Valence electrons are attracted to the positively charged nucleus BUT valence electrons are repelled by the core electrons
- Note that electrons in the same shell "feel" the same attraction to the nucleus (since they are the same distance from the nucleus, just like how the people in the same row would feel the same connection to the artist)

$$Z_{eff} = Z - S$$

Z_{eff}** is the **effective nuclear charge
 ***Z** is the **atomic number** (# of protons)*
 S** is the # of **shielding electrons

Are the shielding electrons core electrons or valence electrons? _____

What would be the effective nuclear charge for Lithium?

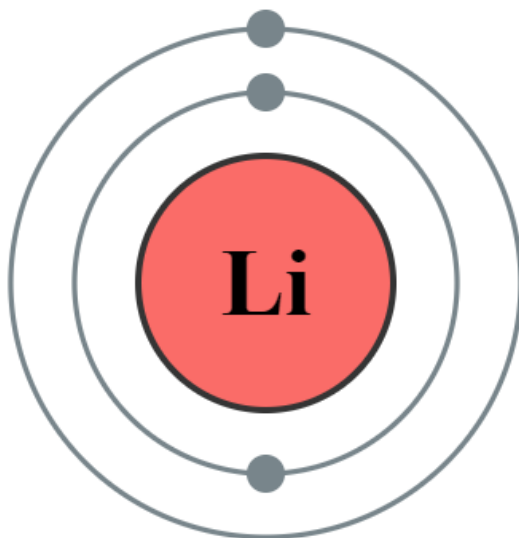


Photo by Greg Robson / CC BY

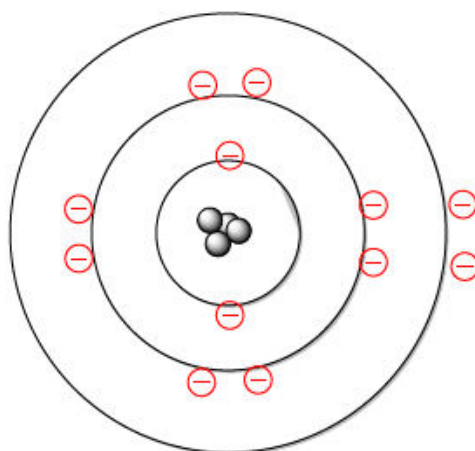
The Periodic Trend

- As we **move to the right** across the periodic table, the # of core electrons stays the same but the **# of protons increases**. Therefore, **Z_{eff} increases**.
 - With $\rightarrow$ protons, can pull the electrons in closer $\rightarrow Z_{\text{eff}}$
 - Effects of shielding are less with more protons
- As we **move down a group** of the periodic table, the **# of core shells increases** and the valence electrons get further from the nucleus. This increased distance from the nucleus leads to a **smaller Z_{eff}** .
 - a group $\rightarrow$ shells
 - shells $\rightarrow$ shielding
 - shielding $\rightarrow Z_{\text{eff}}$

hydrogen 1 H 1.00794																	helium 2 He 4.002602						
lithium 3 Li 6.941	beryllium 4 Be 9.012182																	neon 10 Ne 20.1797					
sodium 11 Na 22.989769	magnesium 12 Mg 24.304																	argon 18 Ar 39.948					
potassium 19 K 39.0983	calcium 20 Ca 40.078	scandium 21 Sc 44.955912	titanium 22 Ti 47.867	vanadium 23 V 50.9415	chromium 24 Cr 51.9961	manganese 25 Mn 54.938045	iron 26 Fe 55.845	cobalt 27 Co 58.933195	nickel 28 Ni 58.6934	copper 29 Cu 63.546	zinc 30 Zn 65.38	gallium 31 Ga 69.723	germanium 32 Ge 72.64	arsenic 33 As 74.9216	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80						
rubidium 37 Rb 85.4678	strontium 38 Sr 87.62	ytrium 39 Y 88.90584	zirconium 40 Zr 91.224	niobium 41 Nb 92.90638	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.9055	paladium 46 Pd 106.42	cadmium 48 Cd 112.411	indium 49 In 114.818	tin 50 Sn 118.710	antimony 51 Sb 121.757	tellurium 52 Te 127.60	iodine 53 I 126.905	xenon 54 Xe 131.29							
cesium 55 Cs 132.9054519	barium 56 Ba 137.327	lanthanides		lanthanum 57 La 138.90547	cerium 58 Ce 140.12	praseodymium 59 Pr 140.90765	neodymium 60 Nd 144.242	promethium 61 Pm [145]	samarium 62 Sm 150.36	europtium 63 Eu 151.964	gadolinium 64 Gd 157.25	terbium 65 Tb 158.92532	dysprosium 66 Dy 162.50014	holmium 67 Ho 164.93032	erbium 68 Er 167.259	thulium 69 Tm 168.93032	ytterbium 70 Yb 173.054						
francium 87 Fr [223]	radium 88 Ra [226]	actinides		actinium 89 Ac [227]	thorium 90 Th 232.0377	protactinium 91 Pa 231.036889	uranium 92 U 238.02891	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	lawrencium 99 Lr [260]									
																		roentgenium 111 Rg [281]					
																		unbinilium 112 Uub [285]					
																		ununilium 114 Uuq [289]					

Example: Estimating Z_{eff}

What is the Z_{eff} of an electron in the $n=3$ shell of magnesium (Mg)?



Mg

2.5.3 Practice

Practice: Estimating Z_{eff}

Calculate the approximate effective nuclear charge of Se.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			









2.6 Atomic Radius Trend

2.6.1

Atomic Radius

The atomic radius is the estimated **radius of an atom** (from the nucleus to the outermost valence electrons)

The Periodic Trend

- As we **move to the right** on the periodic table, the **Z_{eff} increases** and this “pulls” the electrons closer to the nucleus which **decreases the radii**.
- As we **move down a group** in the periodic table, the number of **electron shells increases** which makes the **atom radii larger**.
 - group → shells
 - shells → shielding and Z_{eff}
 - With a Z_{eff} → pull on outer electrons, leading to atomic radii

hydrogen 1 H 1.0079																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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Element with the **largest atomic radii** in the periodic table: _____

Element with the **smallest atomic radii** in the periodic table: _____

How do ions change this?

- **Anions:** _____ due to increased electron-electron repulsion
- **Cations:** _____ due to decreased electron-electron repulsion
 - How does this relate to Z_{eff} ?

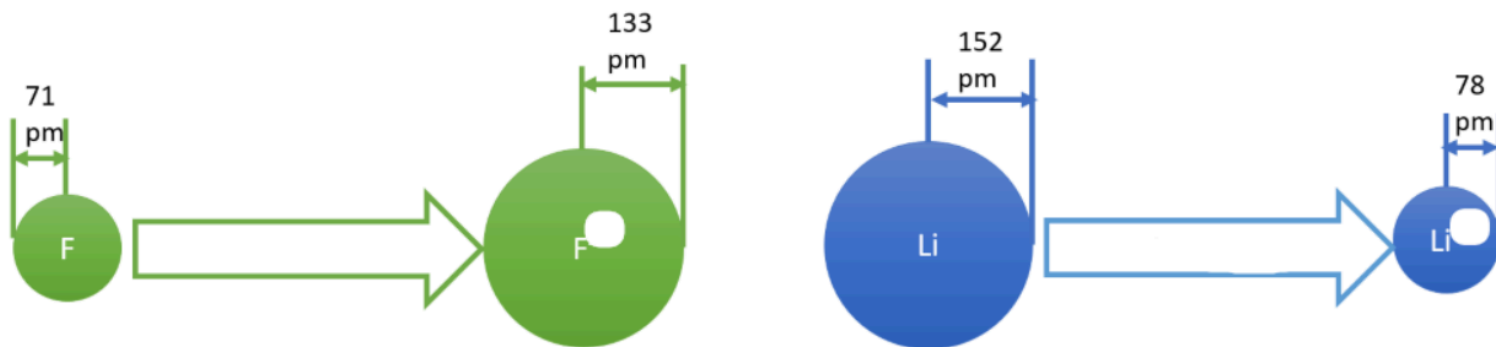
In general:

Ionic Radius: Three Scenarios

You could be asked to rank different atoms or ions according to the sizes of their ionic radii using the trends we discussed.

Here are three common scenarios you may encounter:

1. Same element different charge:



2. Different element same charge:

- Identical trend to atoms

Example: $\text{Li} < \text{Na}$ so $\text{Li}^+ < \text{Na}^+$

3. Different element different charge

- Can only be assessed for **isoelectronic species** (same # of electrons)
- Compare the **proton to electron ratio**.
 - **More protons than electrons**, means stronger pull, **smaller radius**.
 - **Less protons than electrons**, means weaker pull, **larger radius**.

Example: Compare the radii for the following: O^{2-} F^- Ne Na^+ Mg^{2+}

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Example: Ranking Size of Atoms

Rank the following atoms in order of increasing atomic radius: Se, Cs, Br, Ga, F, As.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

Example: Ranking Size of Ions

Rank the following species in order of decreasing size. F^- , N^{3-} , Al^{3+}

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

i WIZE TIP

When comparing atoms that are **isoelectronic** (i.e. they have the same number of electrons), remember **more protons in the nucleus** means a stronger the pull of electrons toward the + charged nucleus, so the **smaller the atomic radius**

Practice: Determining the Largest Atom

Select the atom below which has the largest atomic radius?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

S



I



O



In



Rb



2.7 Ionization Energy

2.7.1

Ionization Energy (IE)

The ionization energy is the amount of **energy required to remove the outermost electron** from an atom or ion in the gas phase

- If removing an electron do we get an anion or a cation? _____



- **↑IE** means that the atom/ion requires **more energy** for a valence **electron to be removed** (has a tight pull on electron)

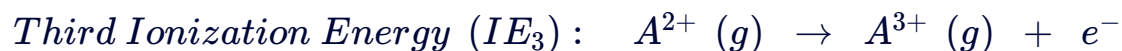
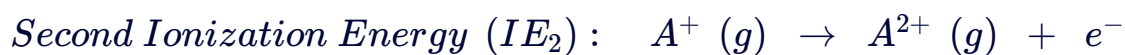
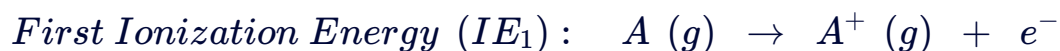
The Periodic Trend

- As we **move to the right** across a period, **Z_{eff} increases** and the electrons are held more tightly. Therefore, it takes more energy to remove an electron and the **IE increases** as we move to the right.
- As we **move down a group** and the valence electrons are further away from the nucleus, they are held more weakly (**lower Z_{eff}**) and **IE decreases** (it becomes easier to remove the outermost electron)

hydrogen																helium																lithium																beryllium																boron																carbon																nitrogen																oxygen																fluorine																neon																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					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Comparing 1st, 2nd, and 3rd Ionization Energies

How does the first IE compare to the second IE?



1st ionization is always smaller than the 2nd because removing a negatively charged electron from a cation is more difficult.

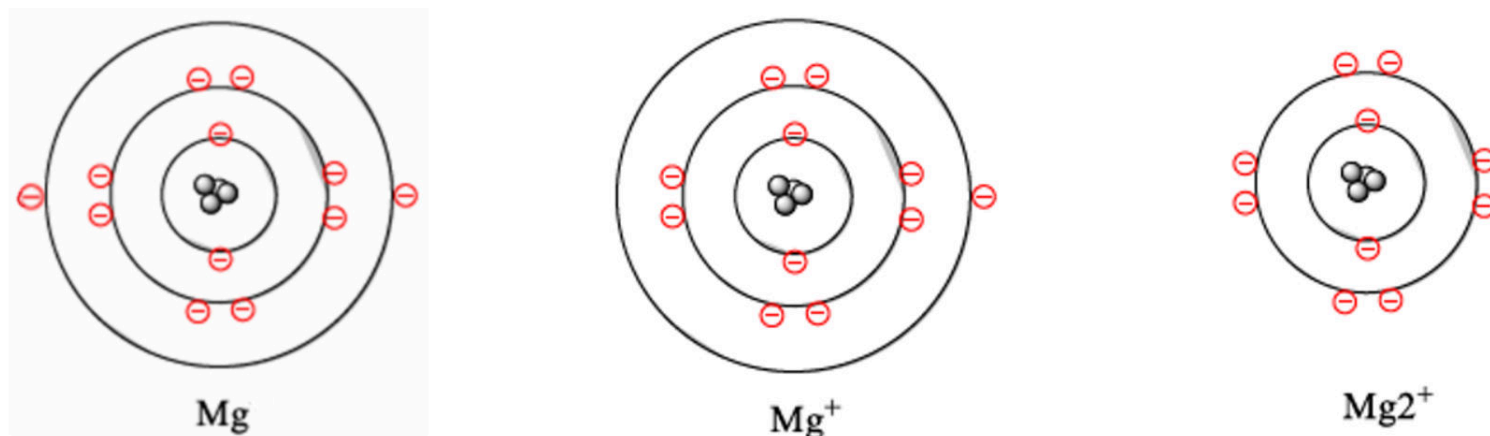
Example:



i WIZE TIP

$$IE_1 < IE_2 < IE_3 < IE_4 \text{ etc}$$

How does the IE_2 and IE_3 for magnesium compare?

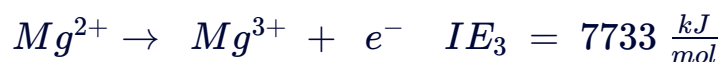
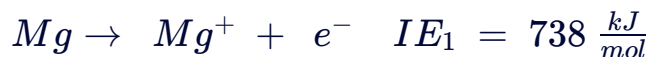


A **core electron is much more difficult to remove than a valence electron**, for example: the 3rd ionization energy of magnesium is much, much, higher than the 2nd.

i WIZE TIP

Although the **IE increases each time another electron is removed, the increase is not linear** and is usually **related to the electron configuration**.

Ionizing Mg^{2+} would be very difficult!



Needing so much energy to remove the 3rd electron indicates that when 2 were removed it was stable/unstable: _____ !

Practice: Highest 1st Ionization Energy

Which of the following species has the highest 1st ionization energy?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

S

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Practice: Trends in Ionization Energy

Rank the following atoms in order of increasing first ionization energy (1 = smallest; 6 = largest): Fe, O, Ba, Si, V, Zr

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

Fe	
O	
Ba	
Si	



V	
Zr	



Practice: Lowest Ionization Energy

Select the element with the lowest 3rd ionization energy.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

A) Sc

☐

B) Mg

☐

C) Cl

☐

D) I

☐

E) C

☐

2.8 Electron Affinity Trend

2.8.1

Electron Affinity (EA)

Electron affinity is the **amount of energy involved with adding an electron** to an atom or ion.

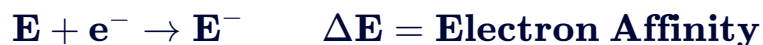
WIZE TIP

Don't confuse ionization energy (IE) with electron affinity (EA)!

- **Ionization energy** is when we **remove** the outermost electron
- **Electron affinity** is when we **add** an electron

Memory tip: Electron affinity has an "a" in it for adding an electron

When adding an electron, do we get a cation or an anion? _____ !



- **If EA is negative** → energy is released/absorbed: _____
 - If this is the case, would the atom go to a higher/lower E state? _____
 - This means the atom has become more/less _____ stable as a result of adding an electron
 - Did this atom like or dislike having an electron added to it? _____
- **If EA is positive** → energy is released/absorbed: _____
 - If this is the case, would the atom go to a higher/lower E state? _____
 - This means the atom has become more/less stable _____ stable as a result of adding an electron
 - Did this atom like or dislike having an electron added to it? _____

Use a **negative EA as a reference point**, unless the question states otherwise.

- ## The Periodic Trend

In general, **as Z_{eff} increases**, the **incoming electron** experiences a greater electrostatic attraction and **stabilization** leading to a **greater (negative) electron affinity**

hydrogen 1 H 1.00794																		helium 2 He 4.002602																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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[293]		ununilium 129 Uuq [293]		unbinilium 130 Uub [293]		ununilium 131 Uuq [293]		unbinilium 132 Uub [293]		ununilium 133 Uuq [293]		unbinilium 134 Uub [293]		ununilium 135 Uuq [293]		unbinilium 136 Uub [293]		ununilium 137 Uuq [293]		unbinilium 138 Uub [293]		ununilium 139 Uuq [293]		unbinilium 140 Uub [293]		ununilium 141 Uuq [293]		unbinilium 142 Uub [293]		ununilium 143 Uuq [293]		unbinilium 144 Uub [293]		ununilium 145 Uuq [293]		unbinilium 146 Uub [293]		ununilium 147 Uuq [293]		unbinilium 148 Uub [293]		ununilium 149 Uuq [293]		unbinilium 150 Uub [293]		ununilium 151 Uuq [293]		unbinilium 152 Uub [293]		ununilium 153 Uuq [293]		unbinilium 154 Uub [293]		ununilium 155 Uuq [293]		unbinilium 156 Uub [293]		ununilium 157 Uuq [293]		unbinilium 158 Uub [293]		ununilium 159 Uuq [293]		unbinilium 160 Uub [293]		ununilium 161 Uuq [293]		unbinilium 162 Uub [293]		ununilium 163 Uuq [293]		unbinilium 164 Uub [293]		ununilium 165 Uuq [293]		unbinilium 166 Uub [293]		ununilium 167 Uuq 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[293]		unbinilium 246 Uub [293]		ununilium 247 Uuq [293]		unbinilium 248 Uub [293]		ununilium 249 Uuq [293]		unbinilium 250 Uub [293]		ununilium 251 Uuq [293]		unbinilium 252 Uub [293]		ununilium 253 Uuq [293]		unbinilium 254 Uub [293]		ununilium 255 Uuq [293]		unbinilium 256 Uub [293]		ununilium 257 Uuq [293]		unbinilium 258 Uub [293]		ununilium 259 Uuq [293]		unbinilium 260 Uub [293]		ununilium 261 Uuq [293]		unbinilium 262 Uub [293]		ununilium 263 Uuq [293]		unbinilium 264 Uub [293]		ununilium 265 Uuq [293]		unbinilium 266 Uub [293]		ununilium 267 Uuq [293]		unbinilium 268 Uub [293]		ununilium 269 Uuq [293]		unbinilium 270 Uub [293]		ununilium 271 Uuq [293]		unbinilium 272 Uub [293]		ununilium 273 Uuq [293]		unbinilium 274 Uub [293]		ununilium 275 Uuq [293]		unbinilium 276 Uub [293]		ununilium 277 Uuq [293]		unbinilium 278 Uub [293]		ununilium 279 Uuq [293]		unbinilium 280 Uub [293]		ununilium 281 Uuq [293]		unbinilium 282 Uub [293]		ununilium 283 Uuq [293]		unbinilium 284 Uub 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[293]		unbinilium 324 Uub [293]		ununilium 325 Uuq [293]		unbinilium 326 Uub [293]		ununilium 327 Uuq [293]		unbinilium 328 Uub [293]		ununilium 329 Uuq [293]		unbinilium 330 Uub [293]		ununilium 331 Uuq [293]		unbinilium 332 Uub [293]		ununilium 333 Uuq [293]		unbinilium 334 Uub [293]		ununilium 335 Uuq [293]		unbinilium 336 Uub [293]		ununilium 337 Uuq [293]		unbinilium 338 Uub [293]		ununilium 339 Uuq [293]		unbinilium 340 Uub [293]		ununilium 341 Uuq [293]		unbinilium 342 Uub [293]		ununilium 343 Uuq [293]		unbinilium 344 Uub [293]		ununilium 345 Uuq [293]		unbinilium 346 Uub [293]		ununilium 347 Uuq [293]		unbinilium 348 Uub [293]		ununilium 349 Uuq [293]		unbinilium 350 Uub [293]		ununilium 351 Uuq [293]		unbinilium 352 Uub [293]		ununilium 353 Uuq [293]		unbinilium 354 Uub [293]		ununilium 355 Uuq [293]		unbinilium 356 Uub [293]		ununilium 357 Uuq [293]		unbinilium 358 Uub [293]		ununilium 359 Uuq [293]		unbinilium 360 Uub [293]		ununilium 361 Uuq [293]		unbinilium 362 Uub 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What about noble gases?

Do you think a noble gas would have a positive or negative electron affinity?

- Are noble gases stable? _____
- Would a noble gas want another electron added? _____
- If you tried to add an electron to a noble gas would that require energy or release energy?

! WATCH OUT!

This trend excludes noble gases. Noble gases have stable, completely filled shells. **Adding electrons to noble gases will break the noble gas configuration.**

Example: Increasing Electron Affinity

Rank the following atoms in increasing absolute energy of their first electron affinity: Br, Ag, Ba, Mo, Sb

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Practice: Positive Electron Affinity

For most elements, the first electron affinity is negative; however, a few elements have a positive first electron affinity. Which atom has the most positive first electron affinity?

☐ F☐ B☐ Ne☐ C☐ O☐ Na

Practice: Highest Electron Affinity

Which of the following has the highest electron affinity?

N

☐

N⁺

☐

N⁻

☐

Not enough information is provided to determine this.

☐

2.9 Electronegativity

2.9.1

Electronegativity (EN)

Electronegativity is a **measure of electron pull**. In a bond, the **more electronegative** element will **pull electrons closer towards itself**

Example: Draw the bond between H and F and show the difference in electronegativity.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	H 2.20																	He
2	Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne
3	Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar
4	K 0.82	Ca 1.00	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.00
5	Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.60
6	Cs 0.79	Ba 0.89	*	Hf 1.3	Ta 1.5	W 2.36	Re 1.9	Os 2.2	Ir 2.20	Pt 2.28	Au 2.54	Hg 2.00	Tl 1.62	Pb 1.87	Bi 2.02	Po 2.0	At 2.2	Rn 2.2
7	Fr 0.7	Ra 0.9	**	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Uut	Fl	Uup	Lv	Uus	Uuo
* Lanthanoids		La 1.1	Ce 1.12	Pr 1.13	Nd 1.14	Pm 1.13	Sm 1.17	Eu 1.2	Gd 1.2	Tb 1.1	Dy 1.22	Ho 1.23	Er 1.24	Tm 1.25	Yb 1.1	Lu 1.27		
** Actinoids		Ac 1.1	Th 1.3	Pa 1.5	U 1.38	Np 1.36	Pu 1.28	Am 1.13	Cm 1.28	Bk 1.3	Cf 1.3	Es 1.3	Fm 1.3	Md 1.3	No 1.3	Lr 1.3		

Electronegativity Values

Photo by DMacks / CC BY

WIZE TIP

General order of EN may come in handy:

$F > O > N > Cl > Br > I > S > C \sim H$

The Periodic Trend

Electronegativity increases going up and to the right in the periodic table.

hydrogen 1 H 1.0079																		helium 2 He 4.0026																			
lithium 3 Li 6.941		beryllium 4 Be 9.0122																boron 5 B 10.81		carbon 6 C 12.011		nitrogen 7 N 14.007		oxygen 8 O 15.999		fluorine 9 F 18.998		neon 10 Ne 20.180									
sodium 11 Na 22.990		magnesium 12 Mg 24.305																aluminum 13 Al 26.981		silicon 14 Si 28.086		phosphorus 15 P 30.974		sulfur 16 S 32.065		chlorine 17 Cl 35.453		argon 18 Ar 39.948									
potassium 19 K 39.098		calcium 20 Ca 40.078		scandium 21 Sc 44.956		titanium 22 Ti 47.867		vanadium 23 V 50.942		chromium 24 Cr 51.996		manganese 25 Mn 54.938		iron 26 Fe 55.845		cobalt 27 Co 58.933		nickel 28 Ni 58.693		copper 29 Cu 63.546		zinc 30 Zn 65.38		gallium 31 Ga 69.723		germanium 32 Ge 72.64		arsenic 33 As 74.922		selenium 34 Se 78.96		bromine 35 Br 79.904		krypton 36 Kr 83.80			
rubidium 37 Rb 85.468		strontium 38 Sr 87.62		yttrium 39 Y 88.906		zirconium 40 Zr 91.224		niobium 41 Nb 92.906		molybdenum 42 Mo 95.94		technetium 43 Tc [98]		ruthenium 44 Ru 101.07		rhodium 45 Rh 102.91		palladium 46 Pd 106.37		silver 47 Ag 107.87		cadmium 48 Cd 112.41		indium 49 In 114.82		tin 50 Sn 118.71		antimony 51 Sb 121.76		tellurium 52 Te 127.60		iodine 53 I 126.90		xenon 54 Xe 131.29			
cesium 55 Cs 132.91		barium 56 Ba 137.33		* 57-70 lanthanide series		lutetium 71 Lu 174.967		hafnium 72 Hf 178.49		tantalum 73 Ta 180.948		tungsten 74 W 183.84		rhenium 75 Re 186.207		osmium 76 Os 190.23		iridium 77 Ir 192.22		platinum 78 Pt 195.08		gold 79 Au 196.967		mercury 80 Hg 200.59		thallium 81 Tl 204.38		lead 82 Pb 207.2		bismuth 83 Bi 208.98		polonium 84 Po [209]		astatine 85 At [210]		radon 86 Rn [222]	
francium 87 Fr [223]		radium 88 Ra [226]		* * 89-102 actinide series		actinium 89 Ac [227]		thorium 90 Th [232]		protactinium 91 Pa [231]		uranium 92 U [238]		neptunium 93 Np [237]		plutonium 94 Pu [244]		americium 95 Am [243]		curium 96 Cm [247]		berkelium 97 Bk [247]		californium 98 Cf [251]		einsteinium 99 Es [252]		fermium 100 Fm [257]		mendelevium 101 Md [258]		nobelium 102 No [259]		lawrencium 103 Lr [262]		unlabeled 104 Uuq [261]	

Which atom is the most electronegative? _____

Example: Increasing Electronegativity

Rank the following atoms in increasing electronegativity: O, Al, Sr, N, Si, Cs

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Practice: Ranking Electronegativity

Choose the option which correctly ranks the following elements in terms of electronegativity.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Ti < Bi < Hg < Cs < Fr

☐

Na < Al < O < S < Cl

☐

Mg < P < C < O < F

☐

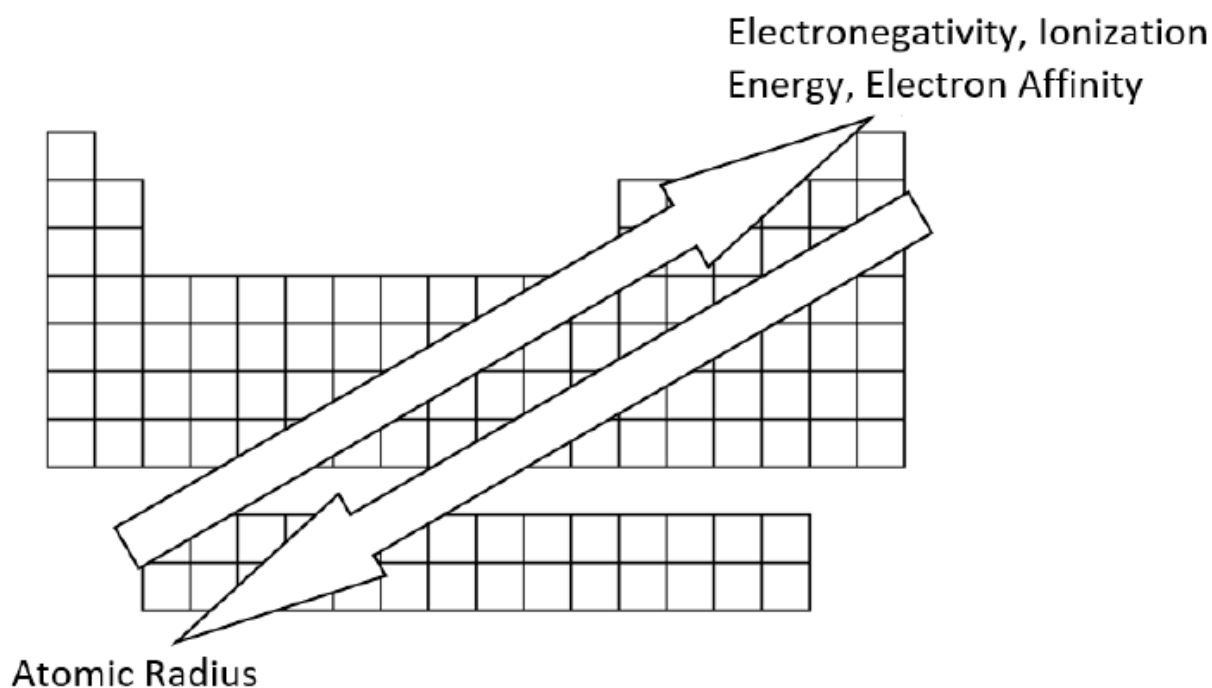
Fe < Co < Mo < F < Cl

☐

2.10 Summary of Periodic Table Trends

2.10.1

Summary of Periodic Trends



Review: which direction does Z_{eff} increase in? _____

Example: Periodic Trends

Label the following statements as either TRUE or FALSE

1. Ionization energy decreases when the atomic size decreases
2. As atomic size increases it gets easier to add an additional electron
3. Mg^{2+} is the same size as Ne since they are isoelectronic

2.

3.

Practice: Periodic Trends

Complete the statements below by filling in the blanks and then choose which option below would help explain your answer.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Part 1

N^{3-} has a (larger/smaller) ionic radius in comparison to Si^{4+} because ...

... of the difference in Z^*

☐

... of the difference in n

☐

... it has a higher proton to electron ratio

☐

... it has a higher electron to proton ratio

☐

Practice: Periodic Trends

Complete the statements below by filling in the blanks and then choose which option below would help explain your answer.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Part 2

Al has a (more negative/ less negative) EA than Si because ...

... of the difference in Z_{eff}

☐

... of the difference in n

☐

... it has a higher proton to electron ratio

☐

... it has a higher electron to proton ratio

☐

Practice: Periodic Trends

Which of the following statements is correct about chlorine and phosphorus atoms?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

- A) The electron affinity of a chlorine atom is smaller than that of a phosphorus atom due to its smaller radius. ☐
- B) A chloride anion has higher electron affinity than a neutral chlorine atom. ☐
- C) A chlorine atom can easily gain and/or lose an electron since it has both a large electron affinity and a large ionization energy. ☐
- D) The effective nuclear charge experienced by the valence electrons in a chlorine atom is smaller than in a phosphorus atom. ☐
- E) All of the above statements are false. ☐

Practice: Attracting Electrons

For an S-O bond, which atom will attract the electrons within the bond more strongly? Why?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Sulfur, because it has a higher electronegativity ☐

Oxygen, because it has a higher electronegativity ☐

Sulfur, because it has a higher Zeff ☐

Oxygen, because it has more valence electrons ☐

Sulfur, because it has more protons ☐

Oxygen, because it is smaller ☐

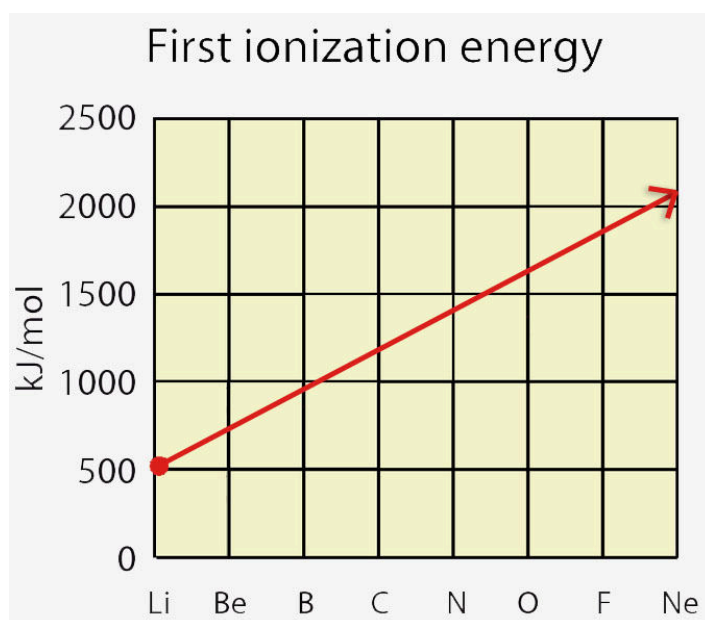
2.11 Exceptions to the Ionization Energy Trend

2.11.1

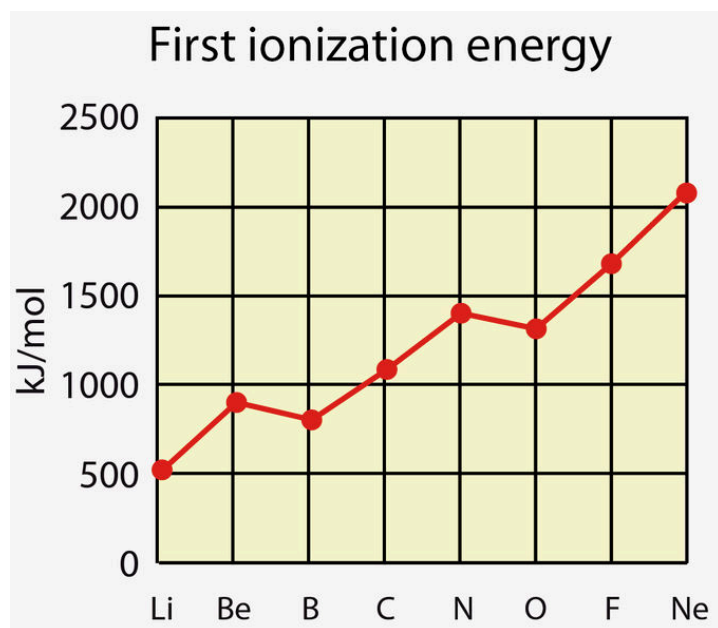
Exceptions to the Periodic Trends Introduction

There are a few exceptions for ionization energy and electron affinity that we should familiarize ourselves with.

Let's consider ionization energy first. **Based on what we currently know**, the trend for ionization energy should look like this:



However, it actually looks something like this:

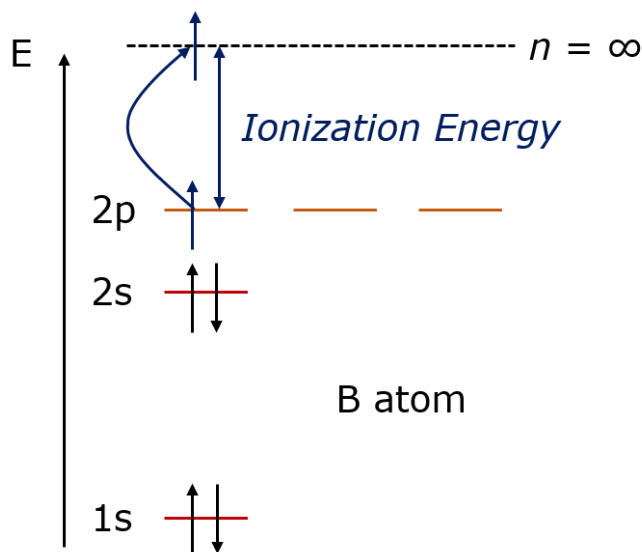
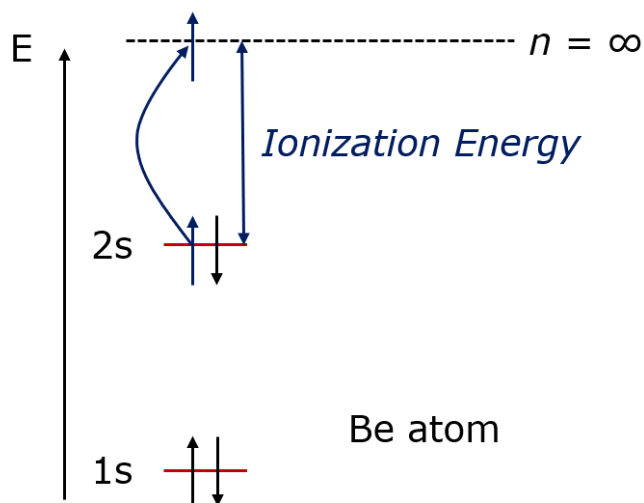


The differences have to do with **electron configurations**. Let's take a look.

Exceptions to the Periodic Trends Examples

Example #1:

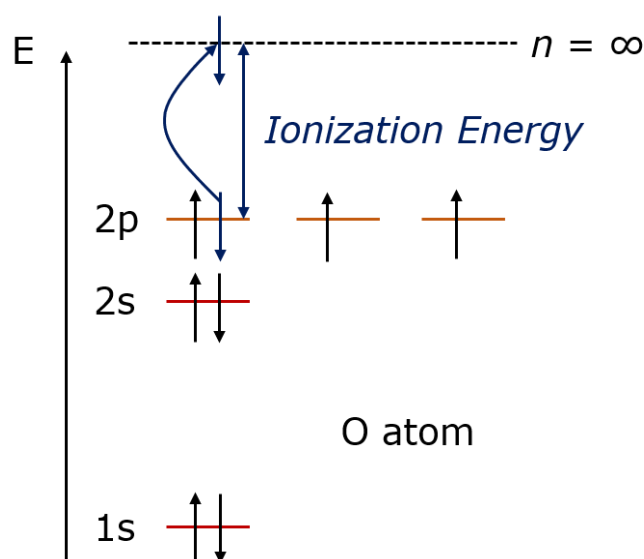
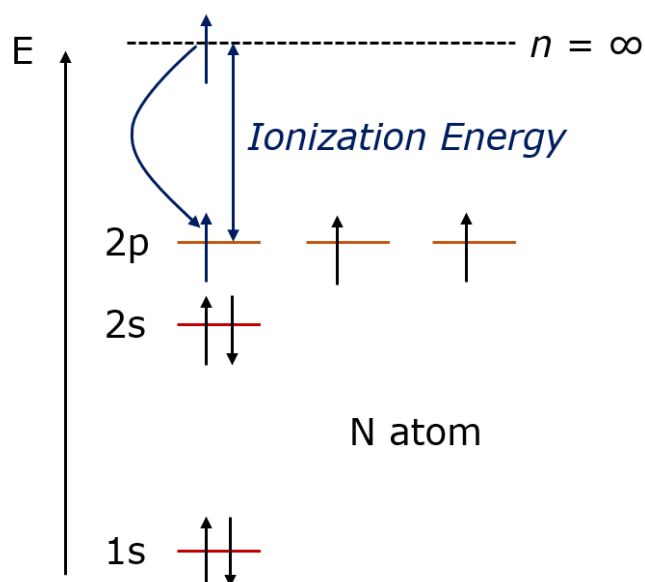
- Based on the periodic trend we learned so far, we would expect that Be or B would have the higher IE: _____



- Recall: which orbital is higher in energy, s or p? _____
- Based on the electron configurations and knowing that **B would LOVE to have its high energy p electron removed** so it can have its **valence electrons in the more stable 2s orbital**, should Be or B have the higher IE? _____

Example #2:

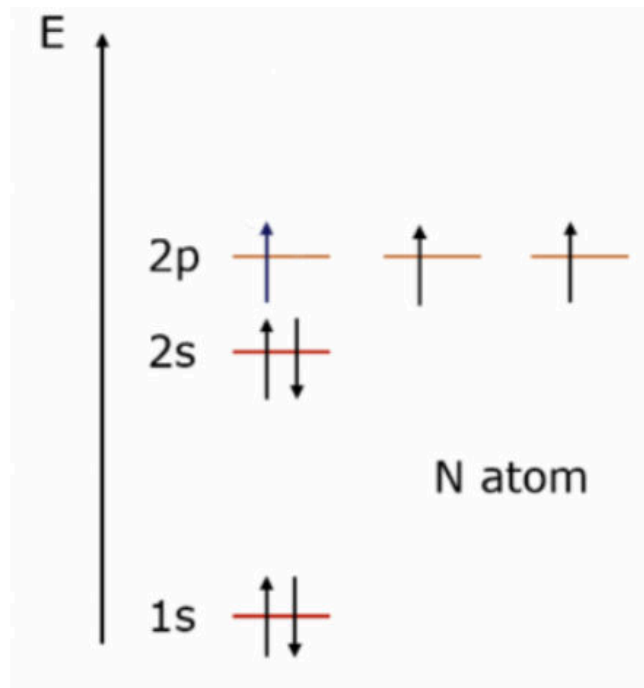
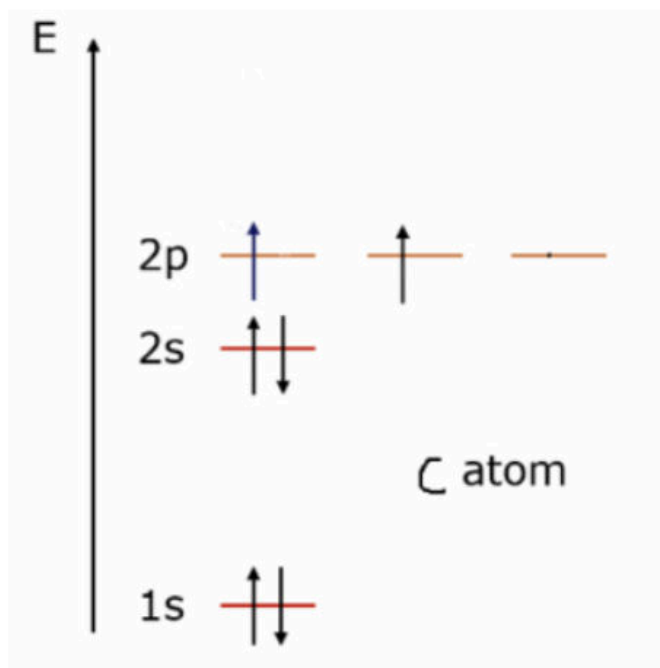
- Based on the periodic trend we learned so far, we would expect N or O to have the higher IE: _____



- Recall: are half-filled subshells more/less stable than partially filled subshells? _____
- Based on the electron configurations and knowing that **N would HATE to have an outermost electron removed** since **it is currently very stable**, should N or O have the higher ionization energy? _____
- In addition, O has some electron-electron repulsion that it would prefer to get rid of. Thus, it makes sense that it has a higher/lower _____ IE

Example #3:

- Based on the periodic trend so far, we would expect C or N to have the higher negative EA: _____



- Based on the electron configurations and knowing that **N would HATE to have an electron added since it is currently very stable with a half-filled 2p subshell**, should N or C have the higher electron affinity? _____

Exceptions to the Periodic Trends Summary

i WIZE TIP

The exceptions specifically happen in **two cases**:

- When we **switch from one type of orbital to another**
Example: 2s to 2p; (Be to B is a jump from 2s to 2p)
- When we **start pairing electrons in a sub-shell**
Example: 2p³ to 2p⁴ (N vs O)

These exceptions can be explained (and remembered) using the **relative energies of atomic orbitals (knowing that the p subshell is > in energy than the s subshell)** and the fact that electrons repel one another and this repulsion is highest for electrons sharing an orbital (**electrons would rather be half-filled in a subshell** instead of having some electrons paired and others unpaired)

Example: Ionization Energy Exceptions

According to ionization energy (IE) trends, you would expect sulfur to have a higher IE than phosphorus; however, the actual ionization data shows the opposite: the IE of sulfur is 1000 kJ/mol and the IE of phosphorus is 1012 kJ/mol. Explain this phenomenon.

Practice: Ionization Energy Exception

Contrary to what we would predict based on Z_{eff} , the ionization energy of aluminum is actually less than magnesium. This is because,

The 3p orbitals of Magnesium are higher in energy than the 3p orbitals of Aluminum ☐

The 3s orbitals of Aluminum are higher in energy than the 3s orbitals of Magnesium ☐

The 3p orbital of Aluminum is higher in energy than the 3s orbital of Magnesium ☐

The 3s orbital of Magnesium is higher in energy than the 3p orbital of Aluminum ☐

The 3p orbital of Magnesium is higher in energy than the 3s orbital of Aluminum ☐

2.12 The Periodic Table & Properties

2.12.1

The Periodic Table of Elements

The **periodic table** is something we are going to see a lot of in chemistry! It organizes the elements by their **atomic number (Z)** and is organized into **groups (columns)** and **periods (rows)**.

Elements in the **same group have very similar reactivity** which we will talk about more when we learn about things like valence electrons, bonding, and Lewis structures.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

You should be familiar with each of the following labels:

- Groups
- Periods
- Alkali Metals
- Alkaline Earth Metals
- Transition Metals
- Nitrogen group (aka Pnictogens)
- Oxygen group (aka Chalcogens)
- Halogens
- Noble (Inert) Gases
- Metals
- Non-metals
- Metalloids
- Lanthanides and Actinides (aka Rare Earth Metals)

Practice: Understanding Noble Gases

Which of the following statements is true?

Noble gases are highly reactive since they want to obtain a full octet.

☐

Noble gases are highly reactive since they want to gain more electrons.

☐

Noble gases are unreactive since they already have a full octet.

☐

Noble gases are very stable because they want to gain more electrons

☐